PYTHON PROGRAMMING



21 Days Challenge

Day-09



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LOOPS AND CONTROL KEYWORDS IN PYTHON: WHILE, FOR, NESTED LOOPS, CONTINUE, AND BREAK

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Table Of Content

- Python For Loop
- Control Statements in Loops
- For-Else Loop
- Nested For Loops
- While Loop

Python Loop

A loop is a way to repeat a block of code multiple times until a certain condition is met.

- **for loop** — repeats code for each item in a sequence.
- **while loop** — repeats code as long as a condition is True

Python For Loop

Syntax of Python for Loop

```
for iterating_var in sequence:
    statement(s)
```

Python For Loop

Python for Loop with Strings

```
string= "Gobeshona"
for i in string:
    print(i, end=" ")
```

Output

```
G o b e s h o n a
```

Python For Loop

Python for Loop with Lists

```
fruits = ["apple", "banana", "cherry"]

for fruit in fruits:
    print(fruit, end=" ")
```

Output

```
apple banana cherry
```

Python For Loop

Python for Loop with Dictionaries

```
fruits = {"apple": "red", "banana": "yellow", "cherry": "red"}

for fruit in fruits:
    print(fruit, "is", fruits[fruit])
```

Output

```
apple is red
banana is yellow
cherry is red
```

Python - For Else Loop

```
for num in range(3):
    print("Number:", num)
else:
    print("Loop is over!")
```

Output

```
Number: 0
Number: 1
Number: 2
Loop is over!
```

Control Statements in Loops

Python break Statement

Python - Continue Statement

Python pass Statement

Python break Statement

break — exit loop immediately

```
for letter in 'Python':
    if letter == 'h':
        break
    print ("Current Letter :", letter)
print ("Good bye!")
```

Output

```
Current Letter : P
Current Letter : y
Current Letter : t
Good bye!
```

Python Continue Statement

continue — skip current iteration, go to next

```
for letter in 'Python':
   if letter == 'h':
      continue
   print ('Current Letter :', letter)
print ("Good bye!")
```

Output

```
Current Letter : P
Current Letter : y
Current Letter : t
Current Letter : o
Current Letter : n
Good bye!
```

Python Pass Statement

pass – do nothing

```
for letter in 'Python':
    if letter == 'h':
       pass

    print ('Current Letter :', letter)
print ("Good bye!")
```

Output

```
Current Letter : P
Current Letter : y
Current Letter : t
Current Letter : h
Current Letter : o
Current Letter : n
Good bye!
```

Python Nested for Loop

```
months = ["jan", "feb"]
days = ["sun", "mon", "tue"]

for x in months:
  for y in days:
    print(x, y)

print("Good bye!")
```

Output

```
jan sun
jan mon
jan tue
feb sun
feb mon
feb tue
Good bye!
```

While Loop

```
a=0
while (a<5):
    a=a+1
    print("hello world")
```

Output

```
hello world
hello world
hello world
hello world
hello world
```

QUESTION SESSION

THANK YOU

https://gobeshona.com.bd/